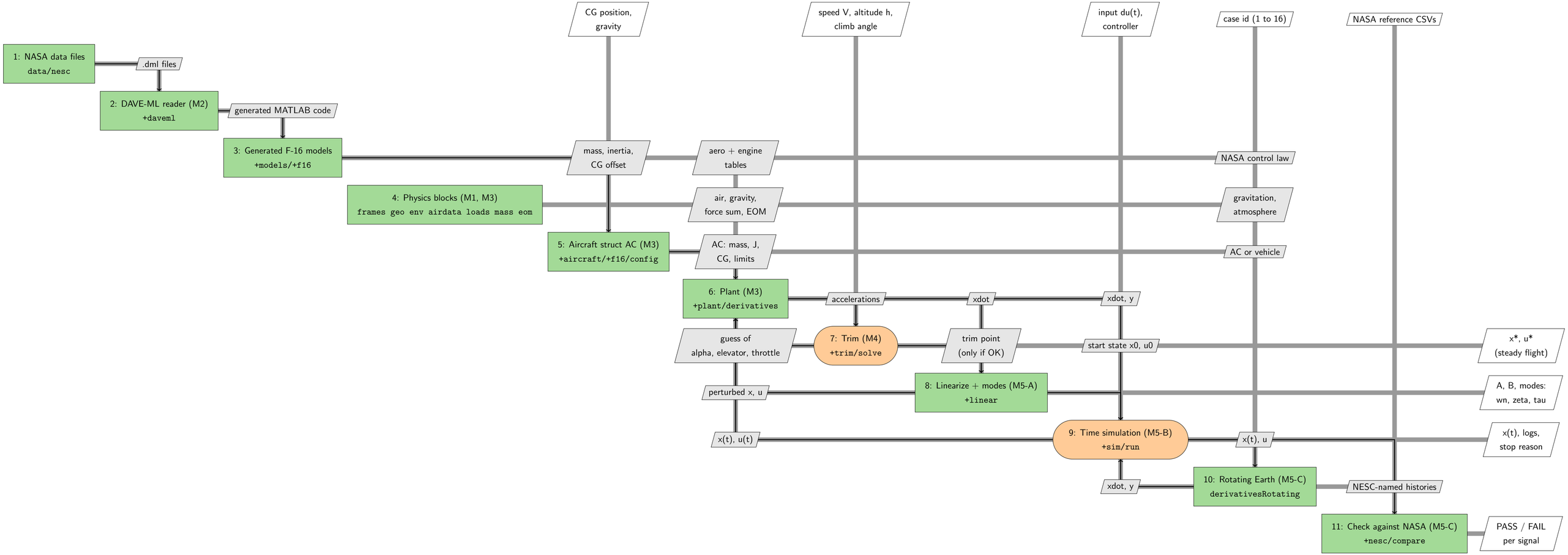
1: NASA data files
data/nesc
.dml files
2: DAVE-ML reader (M2)
+daveml
generated MATLAB code
3: Generated F-16 models
+models/+f16
CG position, gravity
speed V, altitude h, climb angle
input du(t), controller
case id (1 to 16)
NASA reference CSVs
mass, inertia, CG offset
aero + engine tables
NASA control law
4: Physics blocks (M1, M3)
frames geo env airdata loads mass eom
air, gravity, force sum, EOM
gravitation, atmosphere
5: Aircraft struct AC (M3)
+aircraft/+f16/config
AC: mass, J, CG, limits
AC or vehicle
6: Plant (M3)
+plant/derivatives
accelerations
xdot
xdot, y
guess of alpha, elevator, throttle
7: Trim (M4)
+trim/solve
trim point (only if OK)
start state x0, u0
x*, u* (steady flight)
perturbed x, u
8: Linearize + modes (M5-A)
+linear
A, B, modes: wn, zeta, tau
x(t), u(t)
9: Time simulation (M5-B)
+sim/run
x(t), u
x(t), logs, stop reason
xdot, y
10: Rotating Earth (M5-C)
derivativesRotating
NESC-named histories
11: Check against NASA (M5-C)
+nesc/compare
PASS / FAIL per signal